ARGO

Agentic Retrieval & Graph Orchestration for Document Knowledge Systems

ISWC 2026 – Tutorial

25 October 2026

ISWC BARI 2026 · SCAI swiss center for augmented intelligence · iCoSys Institute of Complex Systems · HumanTech Technology for Human Wellbeing Institute · Hes·so · UNI FR Université de Fribourg Universität Freiburg

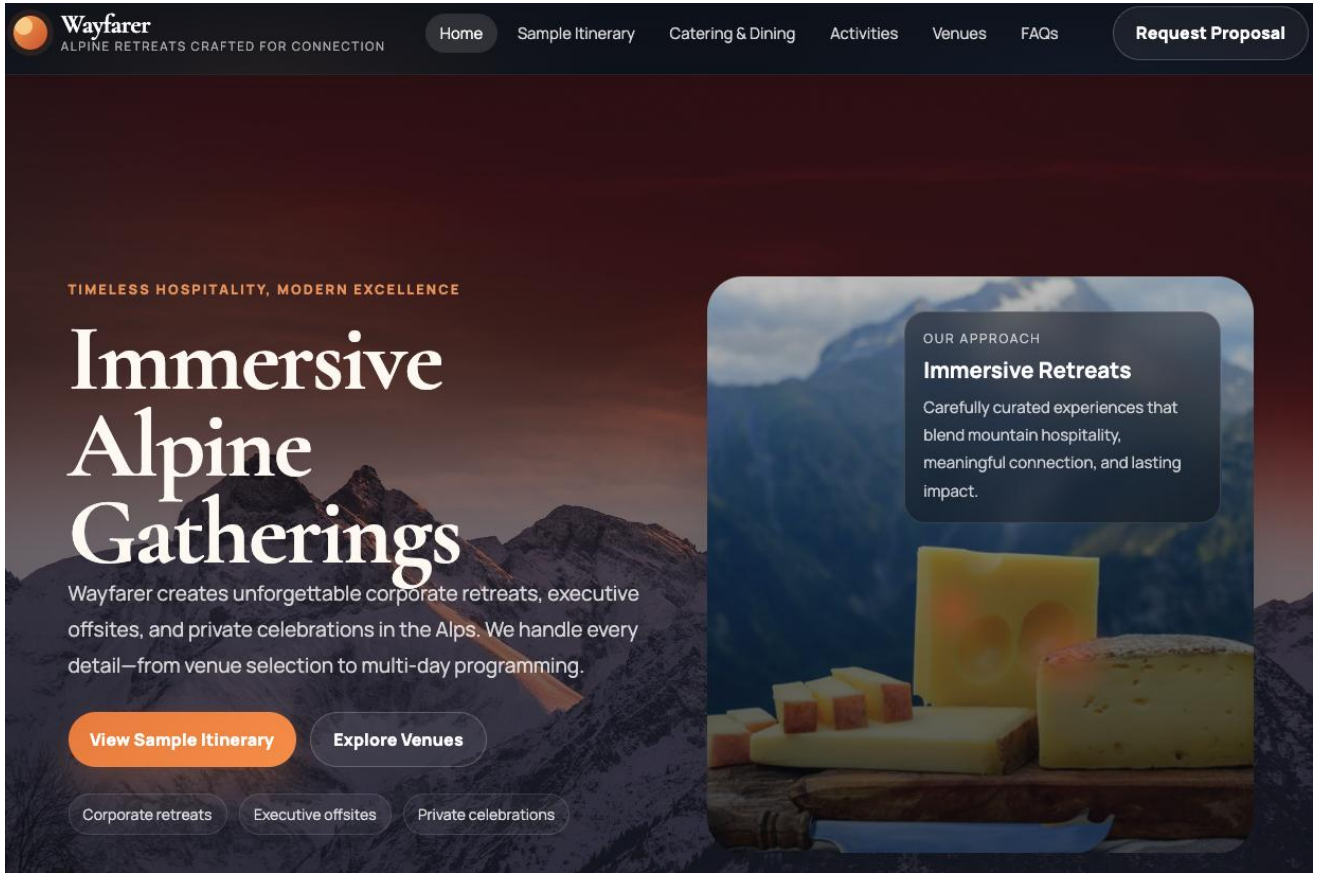
USE CASE

Wayfarer Event Planning Assistant

From business content to evidence-based decisions

Document pack

- **Wayfarer Event Website**
- Email exchange
- Fondue Recipe



https://federicab03.github.io/ARGO_Usecase/

Goal : Build an assistant that helps event organizers evaluate requests and make operational decisions using information from business documents and webpages, supported by clear, traceable evidence.

User request

"Does Wayfarer have a venue that meets the requirements in the email exchange?"

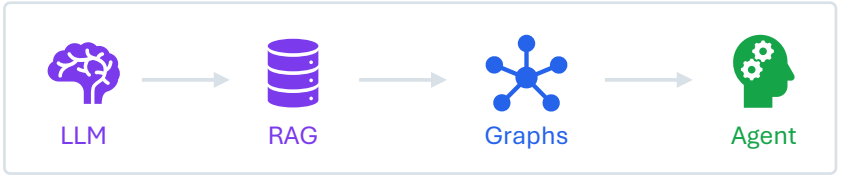
Answer with Evidence:

1) **Decision** (Suitable / Conditional / Not Suitable)

2) **Evidence** (Requirements, venue details, ...)

3) **Source** (Document snippets, emails)

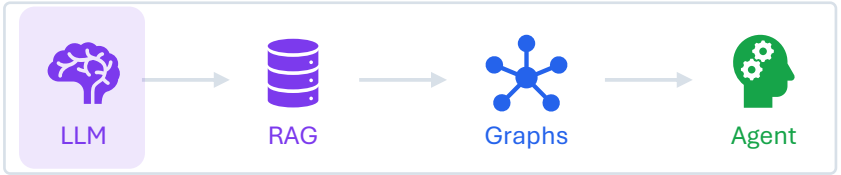
Workshop Timeline



We build the event contract assistant by adding one capability at a time.

Part 01: Foundations	• LLM basics, first PydanticAI app • RAG introduction
Part 02: GraphRAG	• Structural Graph • Sematic Graph
Part 03: Agentic workflow	• Tool calling • Agentic System

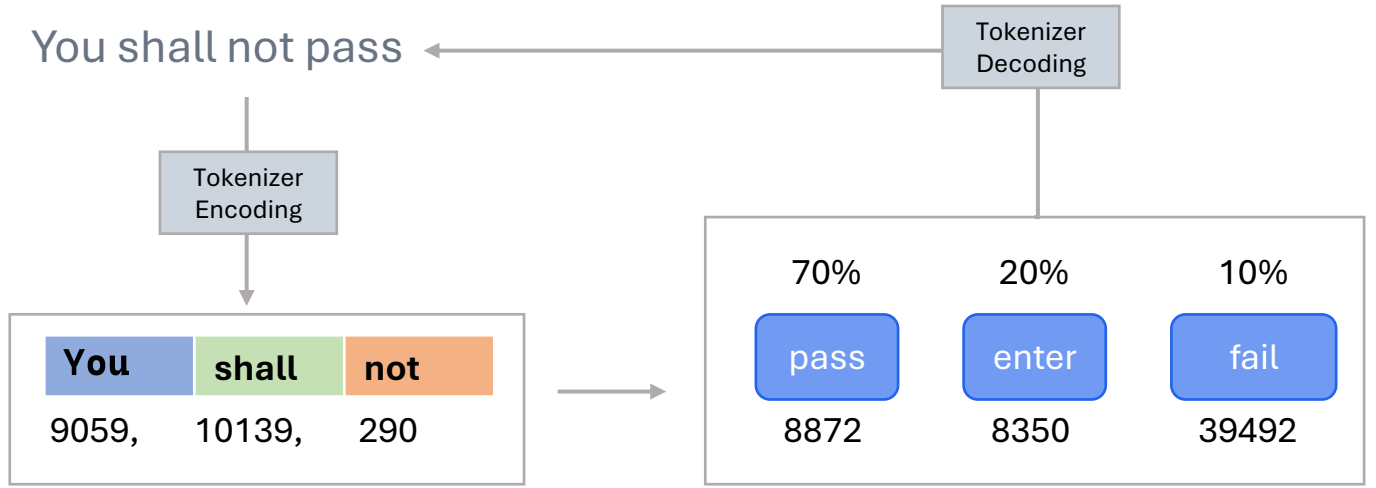
What is an LLM?



AI system trained on vast amounts of text that can understand context and generate human-like language.

Core Idea

1. Context in
2. Tokenize
3. Predict (next-token probabilities)
4. Decode & repeat

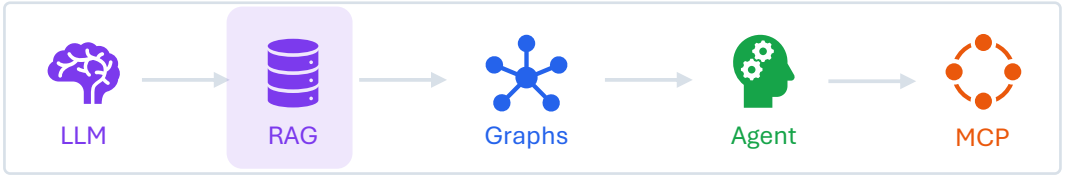


Good at

- Summarization/ Translation
- Question Answering
- Text Generation
- Reasoning over Context

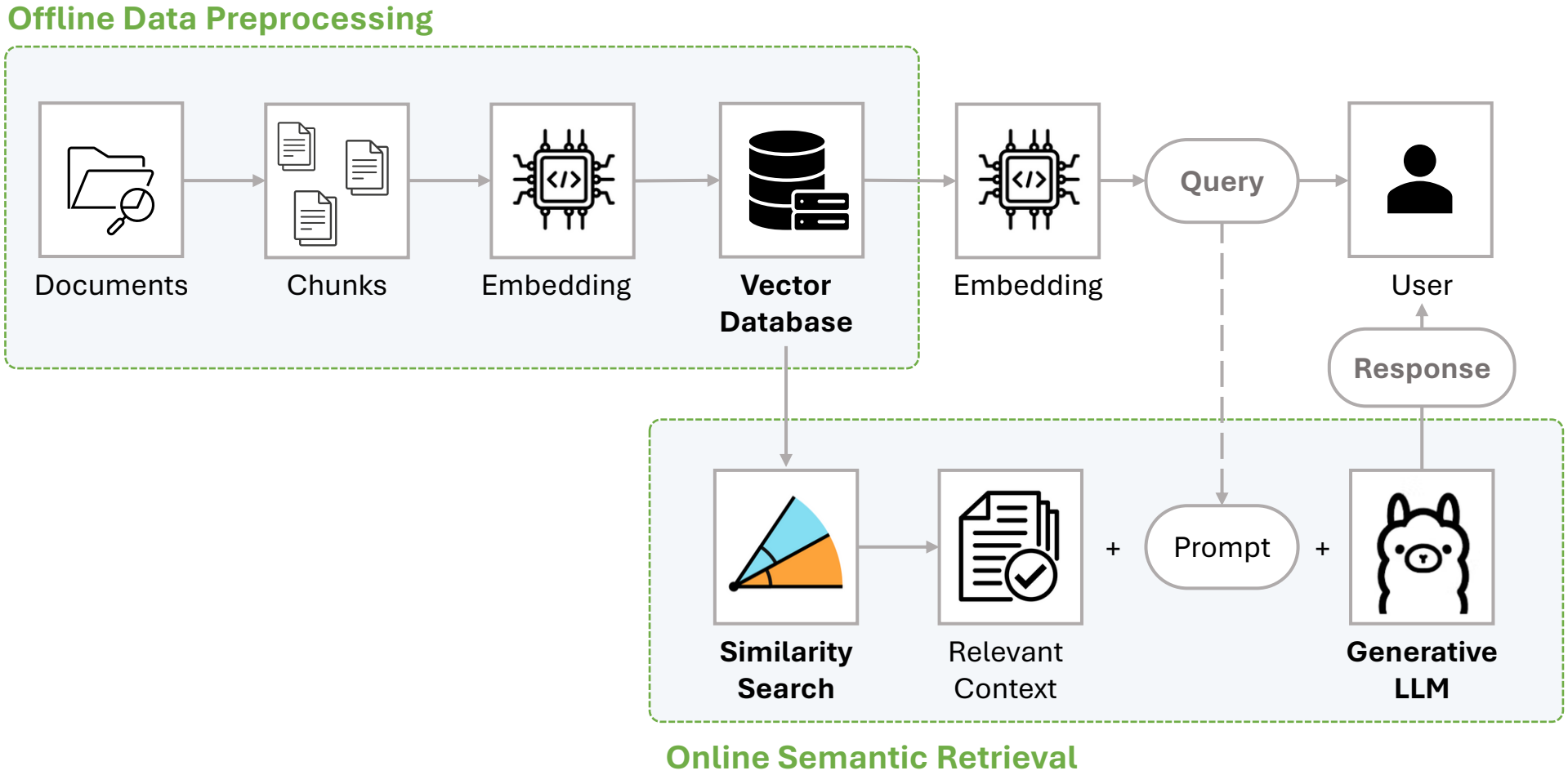
Limits

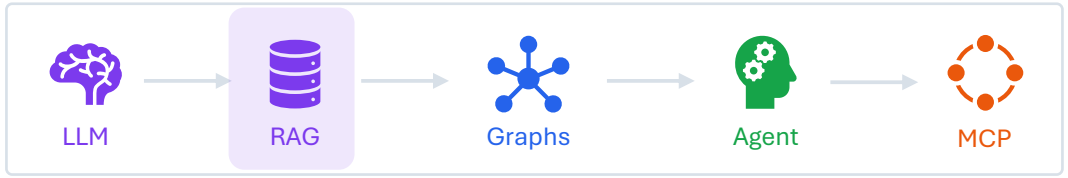
- Hallucinations
- Context Limit
- No native access to external data
- Need tools to act



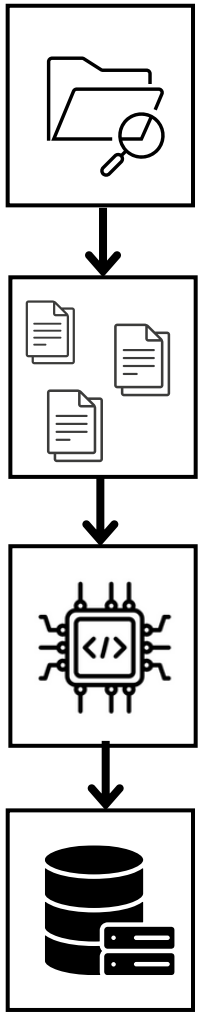
RAG: Retrieval Augmented Generation

The model answers using retrieved document chunks instead of guessing.





RAG: Offline Data Preprocessing



Webpage Extractor

Extract text from webpage

Text Segmentation

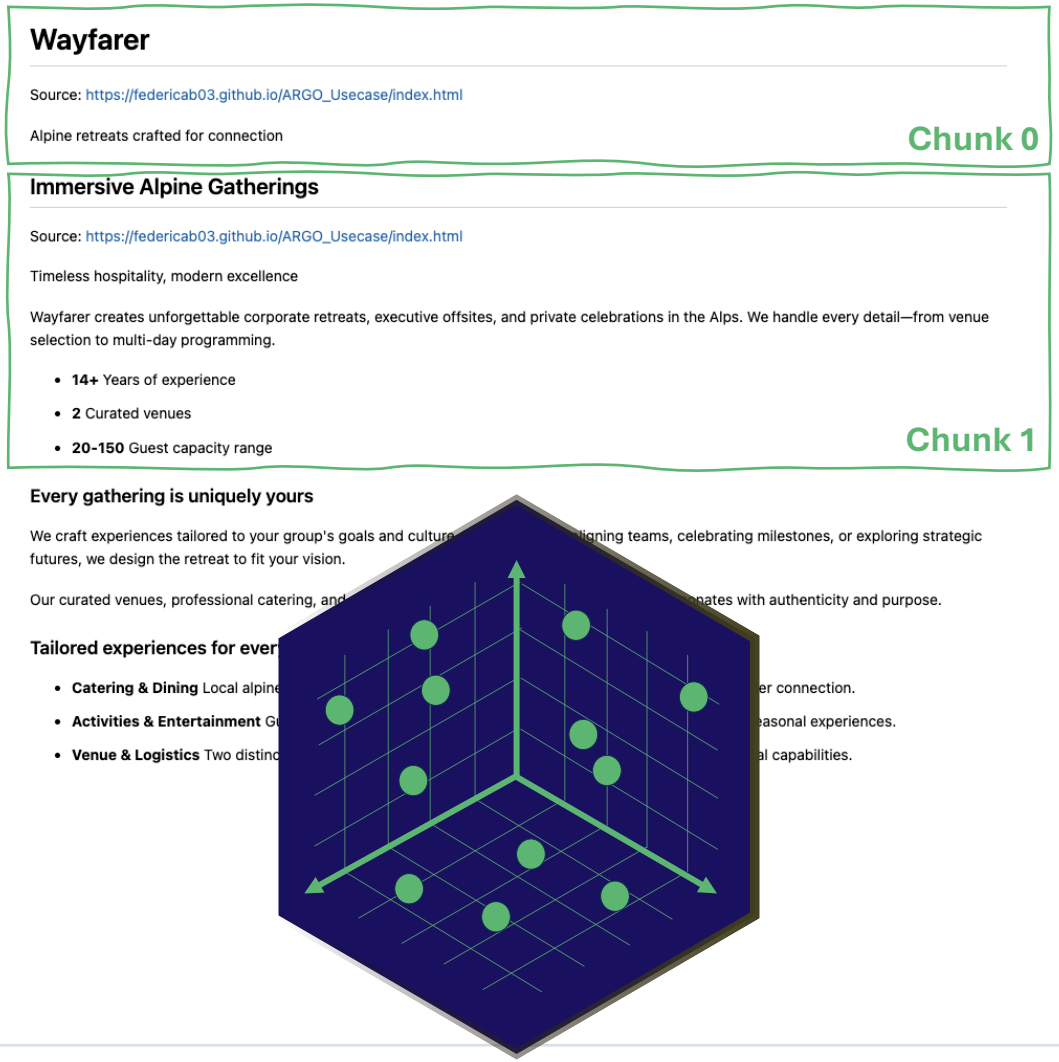
Divide the text in chunks

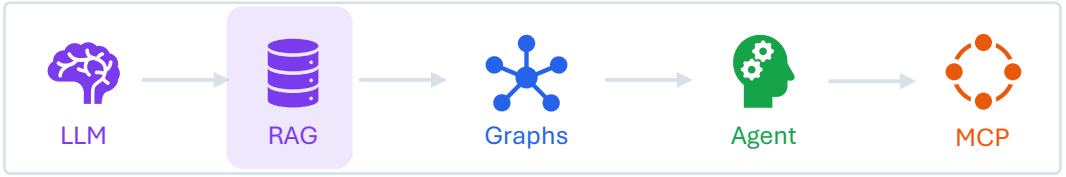
Embedding Model

Transform the chunks in vectors

Vector Database

Store the vectors in a database

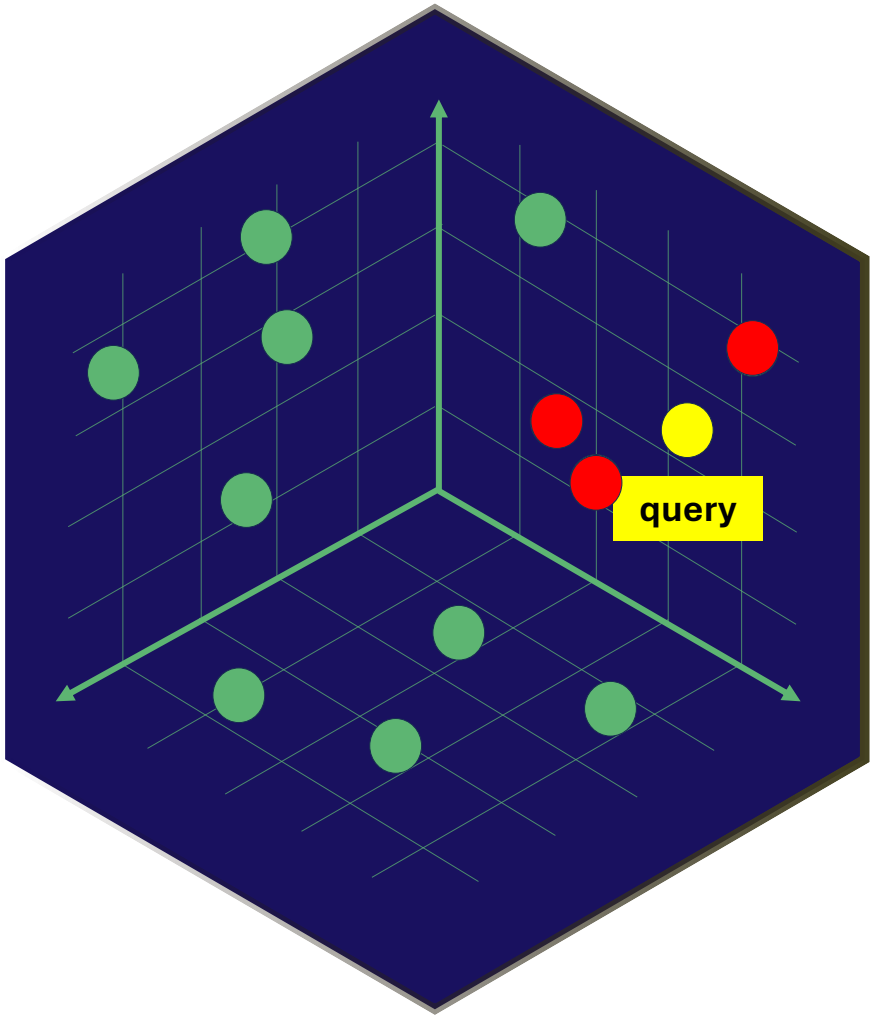




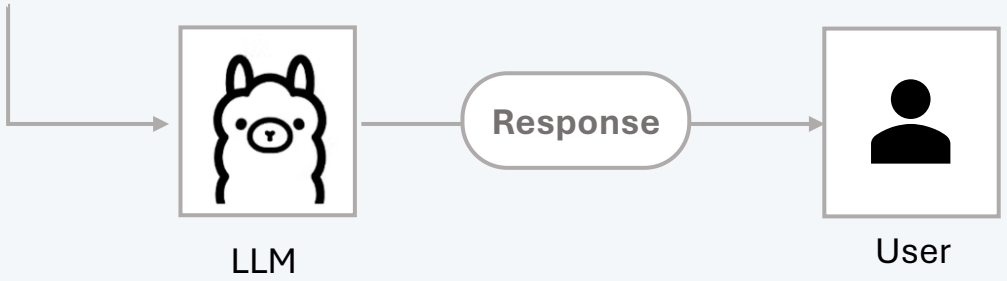
RAG: Online Semantic Retrieval

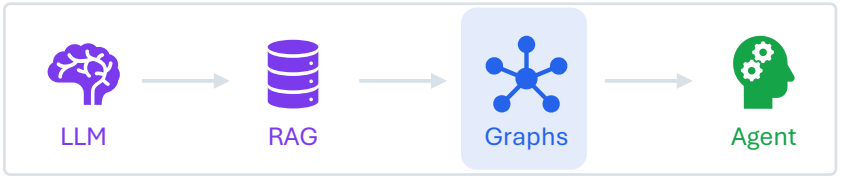
Query : also transform into a vector.
We search the K=3 chunks ”similar” to the query.

- **Metric:** Cosine similarity $\text{Cos_sim}(\mathbf{q}, \mathbf{v}_i) = \frac{\mathbf{q} \cdot \mathbf{v}_i}{\|\mathbf{q}\| \|\mathbf{v}_i\|}$



System Prompt + User prompt + Relevant Context





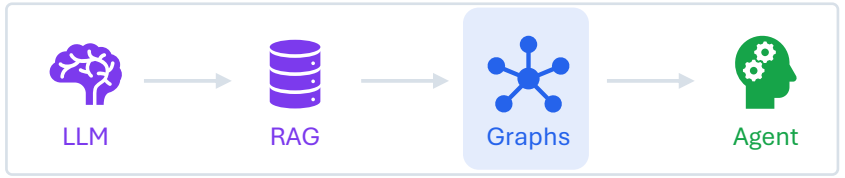
Workshop Repository

Github: https://github.com/argoiswc/argo

```
argo/
├─ data/
├─ exercises/
│   ├─ part_01/
│   ├─ part_02/
│   └─ part_03/
├─ presentation/      ← this presentation
├─ solutions/         ← all the solutions to the exercises
├─ src/               ← utils
└─ README.md
```

Two Graph-RAG implementation options

Structural context expansion or semantic reasoning in Neo4j



OPTION 1

Structural graph + LLM review

Keep the webpage hierarchy, then use the LLM to correct structural relationships.

Webpage chunks → **Structural links** → **LLM relation review** → **Vector seed + 1-hop neighbors**

Best fit

Reliable local context with a simpler pipeline. The graph can run from JSON, with Neo4j as an optional store.

Trade-off: relationships follow document structure, so reasoning stays close to the retrieved section.



Two Graph-RAG implementation options

Structural context expansion or semantic reasoning in Neo4j

OPTION 2

Semantic knowledge graph in Neo4j

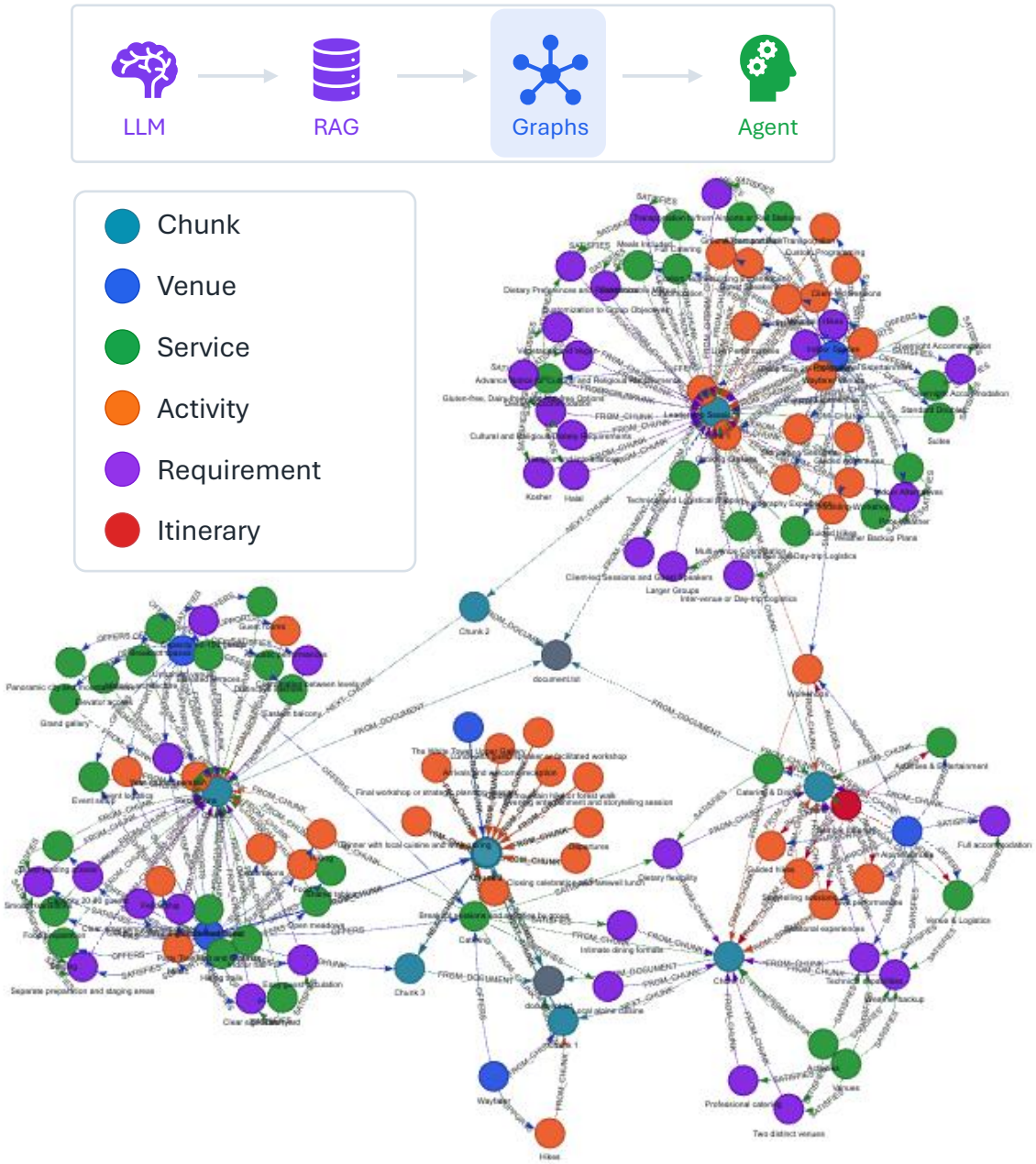
Use an LLM pipeline to extract entities and relationships, then query the graph through Cypher.

Webpage text → **Entity + relation extraction** → **Neo4j knowledge graph** → **Cypher query Graph RAG**

Best fit

Questions that connect venues, constraints, requirements, and other entities across the source content.

Trade-off: richer semantics require Neo4j, an extraction schema, and more validation of generated relationships.





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```

Agent

LLM → RAG → Graphs → Agent

LLM-based system that can reason, choose tools, and act to complete a task.

1

User request

"Is the proposed venue compliant?"

2

Think & plan

Reason about the user needs and the internal knowledge

System prompt:
"You are a helpful assistant. Use the available tools to answer the question accurately and provide supporting evidence."

Context:
(User request, conversation history, uploaded documents...)

3

Act with tools

Chose the right tools

- Text extractor
- OCR extractor
- Webpage retrieval
- Web search

4

Observe & Think

Loop over the user request and the output of the selected tools

Tool results:
Output from the selected tools

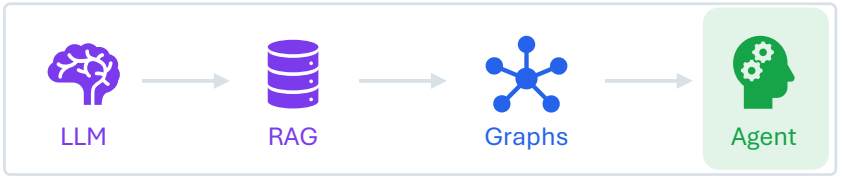
Reasoning:
Evaluate results and decide next step based on the user requests and the updated context

5

Answer with Evidence



Agentic Loop: Thought (Reason about what to do) + **Action** (Use tools) + **Observation** (Reflects on the response from the tool)



Tools

Tools give the agent capabilities beyond the model

Definition

- A tool is a function or service that the agent can call to get information or perform an action.
- The agent decides when to call which tool based on the user's request and available tools.

Good tools

- Have clear, specific name (e.g., get_current_time)
- Have clear description (When to use, what it does, what it returns)
- Have a well-defined input & output schemes (Parameters the model can fill)
- Return structured, useful results (Help the agent to reason)

Think & plan

Based on the user request and available tools the model decides to call a tool

→

Tool calling

The model call the selected tools

→

Tool execution

The system runs the tool and returns the result

→

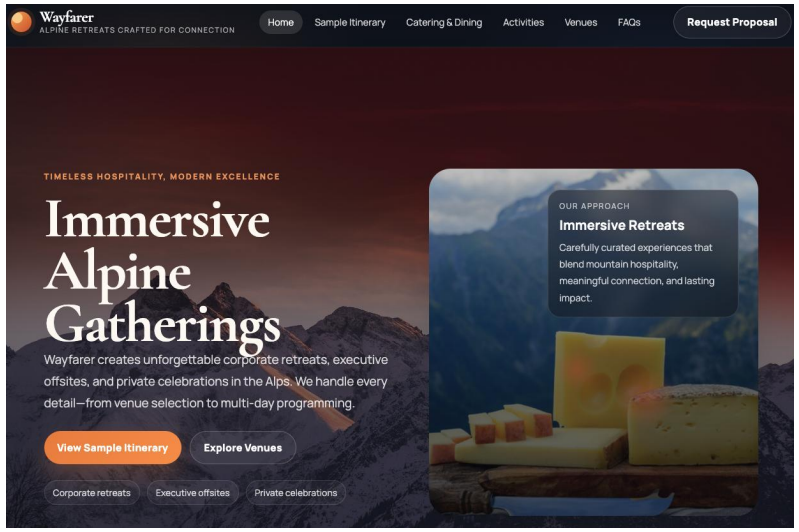
Answer

The results are added to the conversation. The model uses them to produce an answer

Wayfarer Event Planning Assistant

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From business content to evidence-based decisions



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User request

Does Wayfarer have a venue that meets the requirements in the email exchange?

→

Think & plan

Reason about the user needs and the internal knowledge

→

Act with tools

Chose the right tools

→

Answer

Decision + Extracted fields + Cited documents

Goal : Combine extraction, retrieval, and useful tools to support reimbursement decisions

ARGO

Agentic Retrieval & Graph Orchestration for Document Knowledge Systems



Thank you !